

CS & IT ENGINEERING

DISCRETE MATHS
COMBINATORICS



Lecture No. 06



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TOPICS TO BE COVERED

01 Basics of pigeonhole

02 Extended pigeonhole principle

03 Problems in pigeonhole

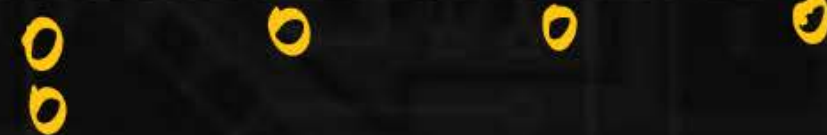
COMBINATORICS



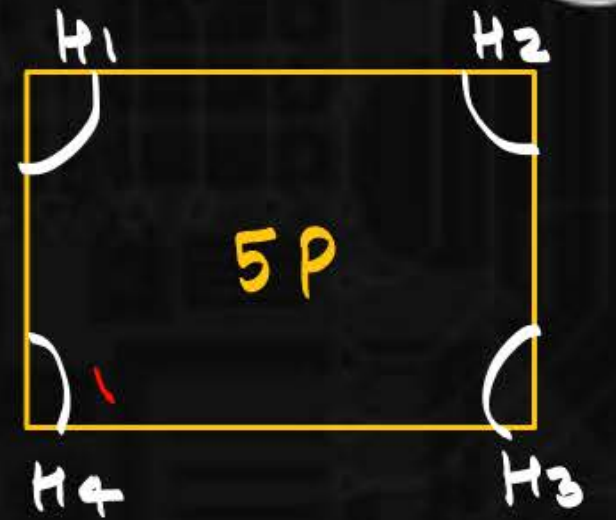
at least 2



H₁ H₂ H₃ H₄



at least 4 · at least 3



COMBINATORICS

Q1. 1500 flights are taking off in single day @ busy airport
 prove that at least 2 flights are taking off in same minute?

<u>at least 2</u>	9.00 m_1	m_2	m_3	...	m_{60}	...	m_{1440}
✓	○	○	○	○	○	...	○
✓	○	○	○	○	○	...	○
✓							

1D
 24 hrs
 $24 \times 60 = 1440$

COMBINATORICS

Q2: In School we have
5 diff. grade system
 what will be min no.
 of students will require
 so that at least 6 students
 will have same grade?

at least 6.

A	B	C	D	E
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0				

Ans: 26.
 26 students.

COMBINATORICS

What will be min no. of students requires such that at least 2 are born in same month?

Ans: 13.

Q2: at least 3. Ans: 25

Q3: at least 4 Ans: 37

Jan	Feb	march	...	Dec
o	o			o
o				
o	o	o	o	o
o	o	o		o
o				

COMBINATORICS



GATE: what will be min no. of
cards will require
so that at least 3
belongs to same suite?

H	S	D	C
0	0	0	0
0	0	0	0
0			

COMBINATORICS



Graph is having 9 vertices and sum of degrees of all vertices are at least 27, prove that some of vertices are having at least 4 degree.

$$\sum d(v_i) \\ = \text{even}$$

28, 30, 32.

v_1	v_2	v_3	...	v_9
0	0			0
0	0			0
0	0			0
0				

$$\begin{aligned}
 &9 \times 3 \text{ degrees} \\
 &= \underline{27 \text{ degrees}} \\
 &+ 1 \\
 &= \textcircled{28}
 \end{aligned}$$

COMBINATORICS



bag contains

6 Red balls

8 blue

10 Green

15 white

20 Yello

what is min. no. of
balls we have
to choose randomly
at least 6 balls of
same color?

Red	blue	Green	White	Yell
6	8	10	15	20



$$5 \times 5 + 1 = 26. \quad \text{Ans: } \underline{26.}$$

COMBINATORICS



$$5 \times 6 + 8 + 1 = \underline{39}$$

bag contains

6 Red balls

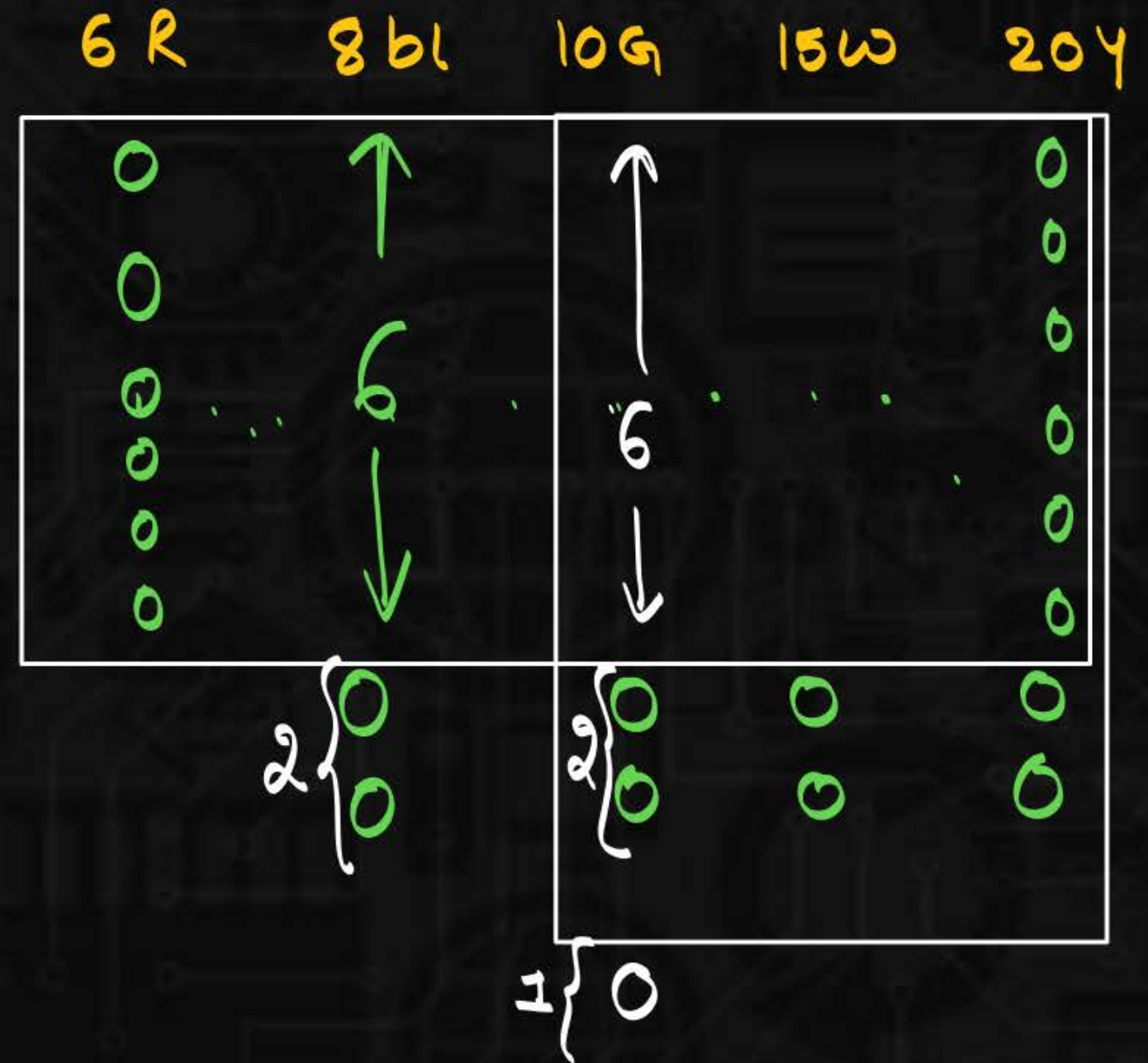
8 blue

10 Green

15 white

20 yellow

what is min. no. of balls we have to choose randomly at least 9 balls of same color?



COMBINATORICS



$$S = \{0 \dots 9\} \quad \begin{array}{l} A \subseteq S \\ |A| = k. \end{array}$$

min size of k .?

any subset of S of size k . contains two distinct

subset of size two. $\{x_1, y_1\}$ $\{x_2, y_2\}$

$$x_1 + y_1 = x_2 + y_2 = 9.$$

$$S = \{ \boxed{0, 1, 2, 3}, \cancel{4}, \cancel{5}, \cancel{6}, \cancel{7}, \cancel{8}, \cancel{9} \}$$

$A \subseteq S$

subset $A = \{ \underline{9, 8, 7, 6, 5, 0, 3} \}$

$\{ \underline{9, 1, 2, 7} \}$ Ans: 4

Student: $\{ \underline{1, 2, 3, 4} \}$

9 8 7 6 5

- Subset 1 $\{ x_1, y_1 \}$ $x_1 + y_1 = 9$ $\{ 0, 9 \} = 0 + 9 = 9$
- Subset 2 $\{ x_2, y_2 \}$ $x_2 + y_2 = 9$ $\{ 6, 3 \} = 6 + 3 = 9$

COMBINATORICS



$$S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$$|K| = 4$$

↓
subset $K = \{1, 8, 2, 7\}$

$$\{x_1, y_1\}$$

$$\{1, 8\}$$

$$\{x_2, y_2\}$$

$$\{2, 7\}$$

$$x_1 + y_1 = x_2 + y_2$$

$$1 + 8 = 2 + 7 = 9$$

COMBINATORICS



0, 2, 3

$$\{ 9 \quad 8 \quad \left\{ \begin{matrix} 7 \\ 2 \end{matrix} \right\} \left\{ \begin{matrix} 6 \\ 3 \end{matrix} \right\} \quad 5 \quad \}$$

$n_1 y_1 \quad n_2 y_2$

$$5 + 2 = 7$$

Ans: 7

COMBINATORICS



box contains 12 Red, 7 blue, n Green balls.

if minimum no. of balls we have to choose

randomly from the box guarantee that 6 balls of

Same color is 15

then n —

COMBINATORICS



hw

Show that if 14 integers are selected from the set
 $S = \{1, 2, \dots, 25\}$ then at least 2 whose sum is 26.

